

Postmortem: Atlas — Relay Delivery Outage in eu-fra

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Summary

Between 02:11 and 05:58 UTC on 14 November 2025, Relay stopped delivering events to customer endpoints in the eu-fra region. Total duration was three hours and 47 minutes. Approximately 1.9 million events were delayed. No events were lost, and no ordering guarantee was violated. Other regions were unaffected throughout. The incident is referred to internally as Atlas, after the deploy that immediately preceded it and which turned out to have no connection to the cause whatsoever; the name stuck before that was established and was not worth changing afterwards.

Impact

340 tenants had at least one delayed delivery. Of those, 61 experienced delays exceeding two hours. Because Relay retains undelivered events rather than dropping them, every affected event was delivered following recovery, and per-endpoint ordering was preserved throughout because the retry path holds subsequent events behind a failing one by design. Four customers opened support tickets during the window, and two subsequently invoked the service credit clause in their Enterprise agreements.

Timeline

Time (UTC)	Event
02:11	mTLS client certificate for eu-fra delivery workers expires. Workers begin failing to authenticate to the internal queue.
02:34	A customer emails support reporting that no webhooks have arrived for twenty minutes. Support is not staffed at this hour.
02:52	Delivery success rate alert fires. Primary on-call is paged and acknowledges within 4 minutes.
03:05	Responder confirms the outage is region-wide in eu-fra and declares S1. Incident channel opened.
03:40	Investigation focuses on the preceding deploy. Deploy is rolled back with no effect.
04:25	Certificate expiry identified in delivery worker logs after log verbosity is increased.
05:44	Certificate reissued manually and distributed to workers. Connections re-establish.
05:58	Delivery resumes. Backlog begins draining.
06:24	Backlog fully drained. Incident closed.

Table 1. Incident timeline.

Cause

The mTLS client certificate used by eu-fra delivery workers to authenticate to the internal event queue expired at 02:11 UTC. The certificate had been issued manually in November 2023 with a two year validity and was never added to the automated renewal inventory, because at the time of issue that inventory was a spreadsheet maintained by one person and the certificate was created during an unrelated incident response. Delivery workers failed closed on authentication failure, which is the correct and intended behaviour, and retried indefinitely against a connection they could not establish.

Detection

Nobody was paged for 41 minutes after onset. The delivery success rate alert was configured against a five minute rolling window but carried a 40 minute evaluation delay inherited from an alerting platform migration completed in early 2025, during which the delay was introduced to suppress noise from a since-fixed metrics pipeline and then never removed. A customer email at 02:34 UTC is therefore the genuine first signal, arriving eighteen minutes before our own monitoring. This is recorded as a detection failure rather than as a monitoring gap, since the alert existed and fired correctly against a misconfigured threshold.

Resolution

The certificate was reissued manually at 05:44 UTC using the same process by which it had originally been created, and workers recovered as connections re-established without requiring a restart. The backlog drained over the following 26 minutes at approximately three times normal throughput, which is the intended behaviour of the drain path and the first occasion on which it had been exercised at this volume in production.

Actions

#	Action	Owner	Status
1	Inventory all manually issued internal certificates and move to automated renewal	Platform	Complete 2025-11-21
2	Add expiry alerting at 30 days for every internal certificate	Platform	Complete 2025-11-21
3	Reduce alerting evaluation delay from 40 minutes to 4 minutes	Platform	Complete 2025-11-15
4	Add synthetic end-to-end delivery probe per region, running every 60 seconds	Platform	Complete 2025-12-03
5	Quarterly review of the certificate inventory	Platform	Recurring from 2026-01

Table 2. Follow-up actions. Action 4 is what detected the February 2026 backlog incident eleven minutes after onset.